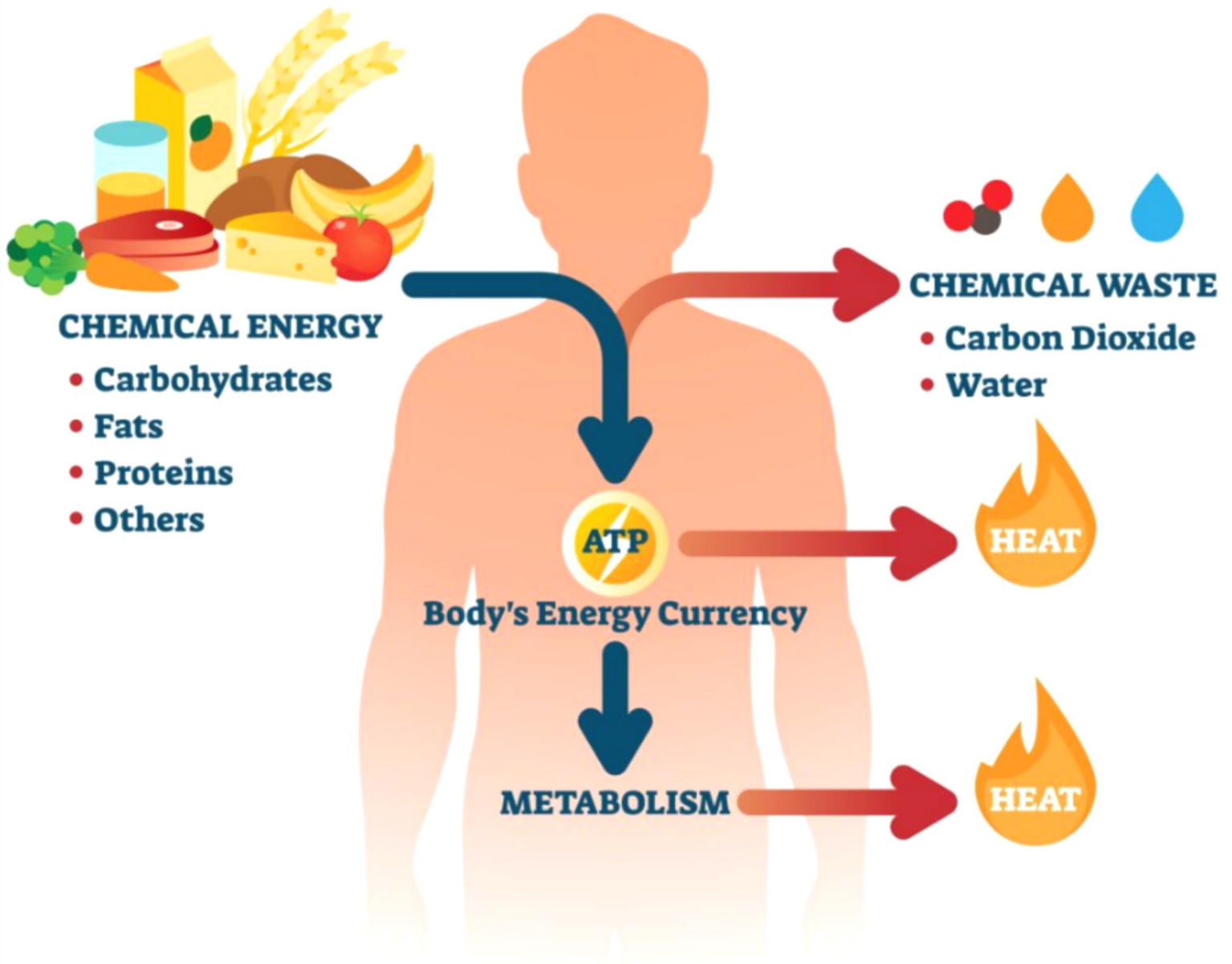


METABOLISM



Nutritional Metabolism

Introduction to Nutritional Metabolism

- Nutritional metabolism is a fundamental aspect of life sciences that explores how the body utilizes nutrients from food to sustain life, support growth, repair tissues, and maintain overall health.
- It bridges the gap between **biochemistry** and human **physiology**, providing insights into the chemical processes that convert food into energy and building blocks necessary for cellular function.
- Nutritional metabolism refers to the intricate series of biochemical reactions and physiological processes by which the body processes nutrients (carbohydrates, proteins, fats, vitamins, minerals, and water) to:

- ◎Generate energy (ATP) required for cellular and systemic activities.

- ◎Synthesize biomolecules essential for growth and repair.

- ◎Regulate biological functions, such as hormonal balance and immune responses.

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- Metabolism, the sum of all chemical reactions in the body, is divided into two opposing yet complementary processes: **catabolism** and **anabolism**.
 - Together, these processes ensure energy balance, growth, and repair, enabling life to sustain itself.

◎ **Catabolism** refers to the breakdown of **complex molecules** into **simpler ones**, resulting in the release of energy. This energy, primarily stored as **adenosine triphosphate** (ATP), is used to power cellular activities.

■ Key Characteristics

- ◎ **Energy-releasing (Exergonic):** Catabolic reactions release energy by breaking chemical bonds in large molecules.
- ◎ **Degradative:** Complex biomolecules (e.g., carbohydrates, lipids, proteins) are degraded into their building blocks.
- ◎ **Oxidative:** Often involves oxidation, where electrons are removed from molecules.
- ◎ **Spontaneous:** Driven by the need to maintain energy homeostasis in the body.

■ Examples of Catabolic Pathways

Glycolysis

Citric Acid Cycle (TCA/Krebs Cycle)

Oxidative Phosphorylation

Lipolysis

Beta-Oxidation

Deamination

Proteolysis

Nucleic Acid Catabolism

■ Biological Importance of Catabolism

- ◎ **Energy Production:** Supplies ATP for cellular processes like muscle contraction, neural activity, and biosynthesis.
- ◎ **Metabolic Intermediates:** Generates precursors for biosynthetic pathways.
- ◎ **Adaptation to Stress:** Provides energy during fasting, exercise, or stress by mobilizing stored energy reserves.

© **Anabolism** is the synthesis of complex molecules from simpler ones, requiring energy input. These reactions build biomolecules essential for growth, repair, and maintenance of the body.

Key Characteristics

- © **Energy-consuming (Endergonic)**: Requires energy input, typically derived from ATP or NADPH.
- © **Bio synthetic**: Constructs large biomolecules such as proteins, nucleic acids, and polysaccharides.
- © **Reductive**: Often involves the addition of electrons (reduction).
- © **Regulated by Hormones**: Insulin, growth hormone, and anabolic steroids often promote anabolic activities.

■ Examples of Anabolic Pathways

Glycogenesis

Gluconeogenesis

Lipogenesis

Cholesterol Biosynthesis

Protein Synthesis

Amino Acid Biosynthesis

Nucleic Acid Anabolism

Collagen synthesis

■ Biological Importance of Anabolism

- © **Tissue Growth and Repair**: Supports cell division, regeneration, and healing processes.
- © **Energy Storage**: Builds energy reserves in the form of glycogen and fat.
- © **Structural Integrity**: Synthesizes structural proteins, lipids, and nucleic acids crucial for cellular and tissue architecture.
- © **Functional Molecules**: Produces enzymes, hormones, and signaling molecules.

■ Interdependence of Catabolism and Anabolism

◎ **Coupled Reactions:** Energy released during catabolism fuels anabolic reactions.

Example: ATP produced in catabolism drives protein synthesis in anabolism.

◎ **Metabolic Cycles:** Intermediates from catabolic pathways (e.g., pyruvate, acetyl-CoA) serve as substrates for anabolic pathways.

■ Biological Significance of Metabolism

Metabolism is critical to maintaining life and is integral to various biological functions:

◎ **Energy Production**

- Carbohydrates, fats, and proteins are metabolized to produce ATP, the primary energy currency of cells, powering processes like muscle contraction, neural transmission, and enzymatic reactions.

◎ **Growth and Repair**

- Anabolic pathways enable the synthesis of cellular components such as DNA, RNA, proteins, and lipids, essential for tissue growth, development, and repair.

◎ **Homeostasis**

- Regulates blood glucose levels, hormonal balance, and electrolyte equilibrium to maintain a stable internal environment.

©Enzyme and Hormone Function

- Nutritional metabolism provides precursors for enzymes and hormones, which are vital for catalyzing reactions and regulating physiological processes (e.g., insulin for glucose regulation).

©Immune System Support

- Nutrients like vitamins A, C, D, and zinc are metabolized to strengthen the immune system, aiding in infection prevention and recovery.

©Detoxification

- Specialized metabolic pathways in the liver detoxify xenobiotics (e.g., drugs, alcohol) and metabolize waste products (e.g., ammonia to urea).

©Adaptation to Stress and Environment

- Metabolic flexibility allows the body to adapt to changes in diet, physical activity, and environmental conditions, such as fasting or prolonged exercise.

©Disease Prevention and Management

- Understanding nutritional metabolism is crucial in preventing and managing metabolic disorders, including diabetes, obesity, cardiovascular diseases, and malnutrition-related conditions.

Thank you!!!

